

Homework 2

Elena Amadei & Min Zhu

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Abstract

In this homework, we perform the 5 exercises provided in the pdf "hw2-2026.pdf".

Exercise 1

During this exercise, given a matrix X , we performed K-means with two different R-vectors. By construction, the distance measure used is the Euclidean distance.

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1.5 & 2 & 0 \\ -0.5 & 0 & -3 \\ 0 & -0.5 & -1 \end{bmatrix}$$

a

The first R-vector is

$$R = \begin{bmatrix} 3 & 3 & 3 \\ -1 & -1 & 0 \end{bmatrix}$$

Firstly, we computed the Euclidean distances between X and R . In Table 1, one can observe the value of the distances and the assignment to the cluster r_1 or r_2 .

$d(X, R)$	r_1	r_2	Closest Centroid
x_1	3.46	3.00	r_2
x_2	3.50	3.91	r_1
x_3	7.57	3.20	r_2
x_4	6.10	1.50	r_2

Table 1: Euclidean distances at first iteration.

Then, we calculated the new centroids performing the average of the X -vectors just assigned. That is $r_1 = x_2$ and $r_2 = \frac{x_1 + x_3 + x_4}{3}$. Therefore, after the first iteration the R-vector looked as follows

$$R = \begin{bmatrix} 1.5 & 2 & 0 \\ 0.17 & 0.17 & -1 \end{bmatrix}$$

Subsequently, we computed again the distances with the new centroids in order to check if the clusters just created could be confirmed. In Table 2, we present our calculations.

$d(X, R)$	r_1	r_2	Closest Centroid
x_1	1.50	2.32	r_1
x_2	0	2.48	r_1
x_3	4.12	2.11	r_2
x_4	3.08	0.69	r_2

Table 2: Euclidean distances at second iteration.

As one can notice, after the second iteration the vector x_1 resulted in being assigned to the cluster r_1 . Consequently, we re-computed the centroids according to the new clusters. The newly computed R-vector can be seen below.

$$R = \begin{bmatrix} 1.25 & 1.5 & 0.5 \\ -0.25 & -0.25 & -2 \end{bmatrix}.$$

At the third iteration of the K-means calculations, none of the X vectors changed cluster. So we confirmed that the vectors x_1 and x_2 belong to the cluster r_1 , and x_3 and x_4 belong to the cluster r_2 . In Table 3, the calculations can be observed.

$d(X,R)$	r_1	r_2	Closest Centroid
x_1	0.75	3.48	r_1
x_2	0.75	3.48	r_1
x_3	4.19	1.06	r_2
x_4	2.80	1.06	r_2

Table 3: Euclidean distances at third iteration.

b

In this section, the R-vector was given as

$$R = \begin{bmatrix} -0.5 & 0 & -3 \\ -1 & -1 & 1 \end{bmatrix}.$$

As before, we firstly computed the distances with the given starting values for the centroids, and we temporary assigned the X vectors to the closest centroid. In Table 4, we display the results.

$d(X,R)$	r_1	r_2	Closest Centroid
x_1	4.38	2.83	r_2
x_2	4.12	4.03	r_2
x_3	0	4.15	r_1
x_4	2.12	2.29	r_1

Table 4: Euclidean distances at first iteration.

Based on this first assignment, we computed the new centroids, so that R was as follows,

$$R = \begin{bmatrix} -0.25 & -0.25 & -2 \\ 1.25 & 1.5 & 0.5 \end{bmatrix}$$

Already at the second iteration, the clusters were confirmed with vectors x_1 and x_2 belonging to r_2 , while x_3 and x_4 belonging to r_1 .

$d(X,R)$	r_1	r_2	Closest Centroid
x_1	3.38	0.75	r_2
x_2	3.48	0.75	r_2
x_3	1.06	4.19	r_1
x_4	1.06	2.79	r_1

Table 5: Euclidean distances at second iteration.

In conclusion, we could observe that the clusters formed in task a and task b coincide. The main difference is that with the R-vector given in task a, it took a larger number of iterations, resulting in *solution b* being the best one.

Exercise 2

In this question, we work on a matrix $A \in R^{16 \times 3}$, where each row represents a node in a three-dimensional space. First, we draw a similarity graph using the ϵ – neighborhood method with settings $\{c = 2, \epsilon = 2.5\}$ and it is visualized in Fig 1.

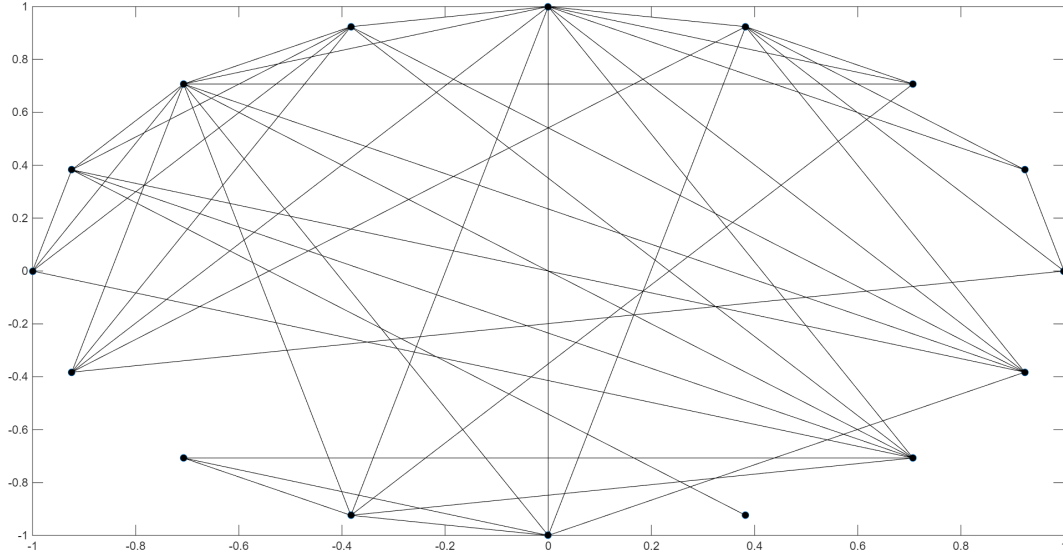


Figure 1: Similarity Graph when $c = 2, \epsilon = 2.5$

a

From the graph, We can see all nodes are connected by edges, resulting in a connected graph with no isolated nodes. Hence, the graph contains exactly 1 connected component.

b

The unnormalized graph Laplacian L is defined as

$$L = D - W$$

where D is the degree matrix and W the weight matrix of the graph.

In our case, the result of $\text{eig}(L)$ is

$$\lambda_1 = \{-0.0000, 0.8188, 1.5428, 2.5207, 3.2215, 3.6453, 4.3127, 5.2325, \\ 5.9444, 6.0494, 7.5214, 7.9123, 8.0162, 8.8315, 11.0997, 11.3307\}$$

There is only one eigenvalue equal to 1, corresponding to the zero eigenvalue $\lambda = 0$ with multiplicity one. Combined with the result in part (a), where the graph has exactly one connected component, this verifies Lemma 2.3.3 in EJ2, which states that for unnormalized Laplacian, the multiplicity of the zero eigenvalue equals the number of connected components.

c

In this subquestion, the value of c is changed to 3, and ϵ remains 2.5, this leads to a new similarity graph shown in Fig.2.

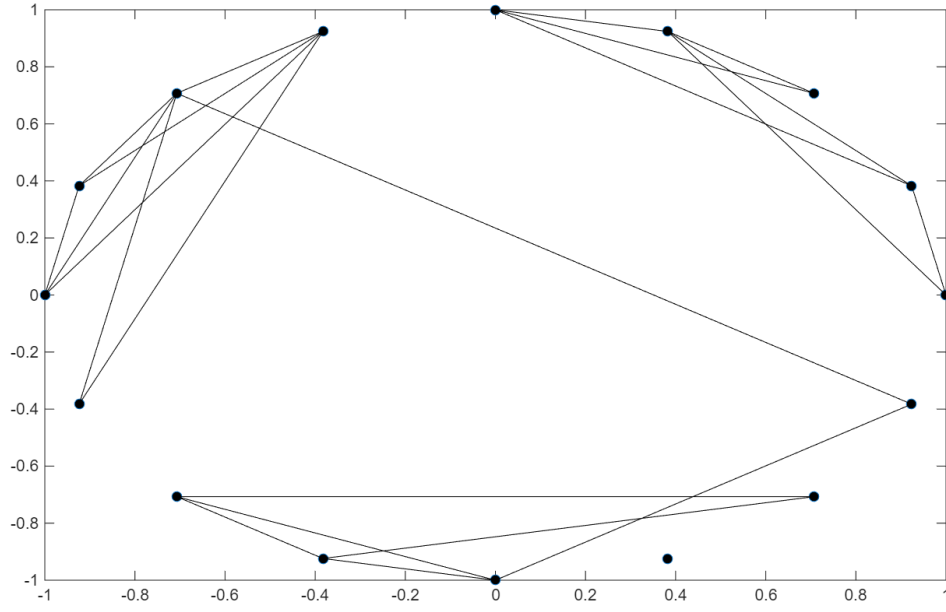


Figure 2: Similarity Graph when $c = 3, \epsilon = 2.5$

Under this setting, the corresponding eigenvalue of the unnormalized Laplacian is

$$\lambda_2 = \{-0.0000, 0.0000, 0.0000, 0.1795, 1.4587, 1.5858, 2.0000, 2.7796, 3.0000, 4.0000, 4.0000, 4.4142, 4.4991, 5.0000, 5.0000, 6.0630\}$$

Fig 2 shows that the graph consists of three connected components. Meanwhile, the eigenvalue λ_2 contains three zero values, indicating that the zero eigenvalue has multiplicity three. This again confirms Lemma 2.3.3.

d

Then, we change the value of c and ϵ to 3.1 and 2.8 respectively, and the similarity graph changes to Fig.3.

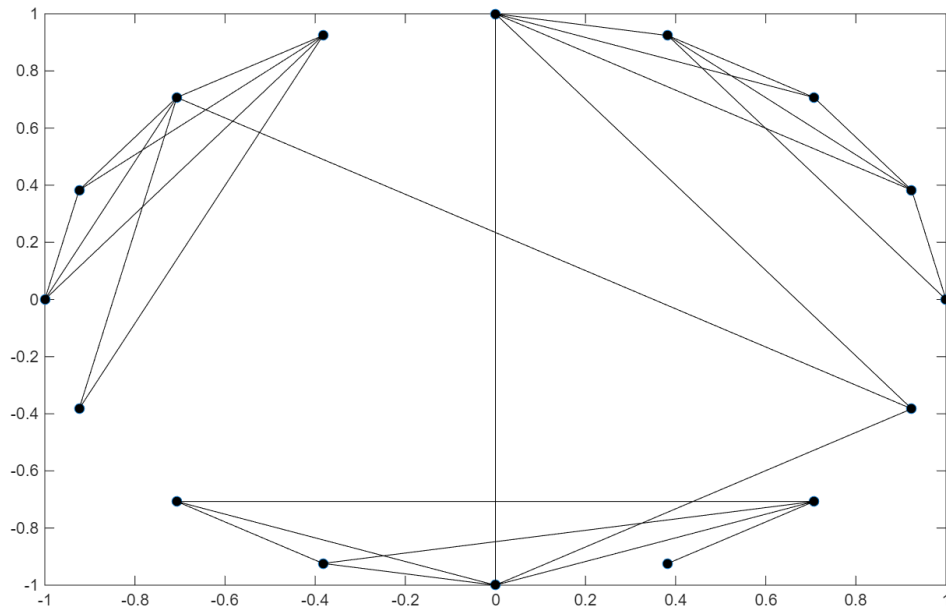


Figure 3: Similarity Graph when $c = 3.1, \epsilon = 2.8$

Extracting the second eigenvalue from the eigenvector and plot it with i , we get the $v_2(i)$ with i as shown in Fig.4.

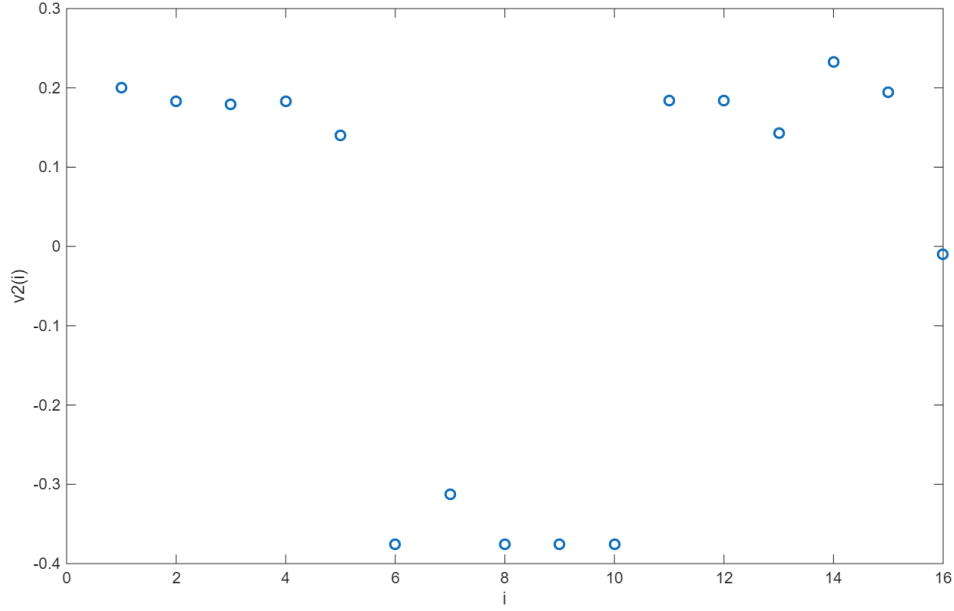


Figure 4: Second eigenvector against i

With the setting of $\{c = 3.1, \epsilon = 2.8\}$, the graph is connected with only one zero eigenvalue. According to the Fiedler's theorem, the sign of $v_2(i)$ can be used for separation. The plot of the second eigenvector against i clearly shows that $v_2(i)$ are separated into distinct groups: nodes 1-5 and nodes 11-15 have the second eigenvalue around $+0.2$, while nodes 6-10 are around -0.4 . Node 16 lies near the origin, suggesting it is a non-representative point that doesn't strongly belong to either main cluster identified by v_2 alone.

e

We continue to plot the third eigenvector v_3 against the second eigenvector v_2 , which can be seen in Fig.5.

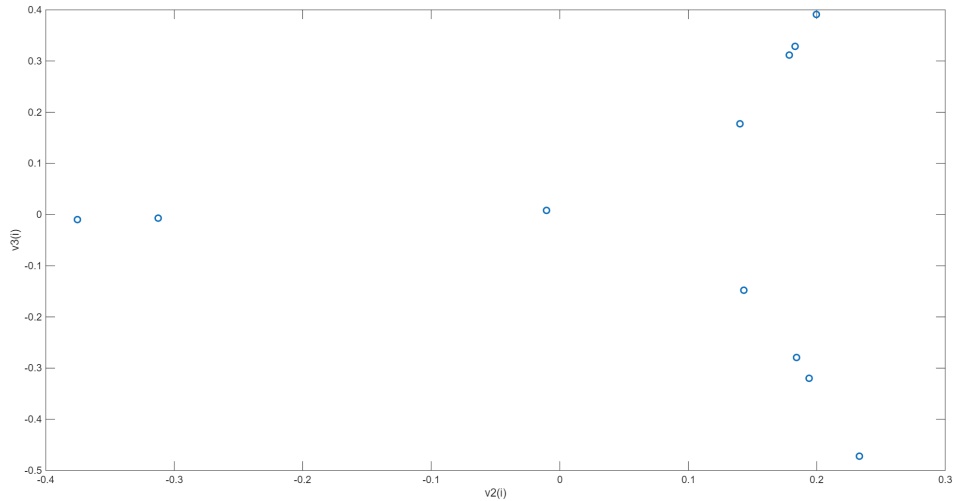


Figure 5: Third eigenvector against second

This plot reflects a finer clustering of the graph nodes compared to Fig 4 because, we can read from the x-axis that v_2 maintains the primary separation of positive and negative groups. What's

more, along the y-axis, v_3 further partitions the positive component into two sub-clusters: one is $v_3 \in (0.2, 0.4)$, and another is $v_3 \in (-0.5, -0.15)$. Therefore, three clusters are read from this plot. Hence, this plot reflects a finer clustering of the graph nodes.

The eigenvalues under this setting is

$$\lambda_2 = \{0.0000, 0.1667, 0.3214, 1.0866, 2.0000, 2.0148, 2.5588, 4.0000, 4.0000, 4.6646, 4.8718, 5.0000, 5.0000, 5.4199, 6.2625, 6.6328\}$$

It is obvious that there exist a huge gap between the third(0.3214) and the fourth(1.0866) value, indicating that there are three clearly-separated clusters, which is align with our conclusion drawn from the plot.

Exercise 3

During this Bengali cleanup simulation, we performed clustering to identify seven regions with similar plastic pollution.

a

For this task, a map with all the measurement stations is displayed in 6.

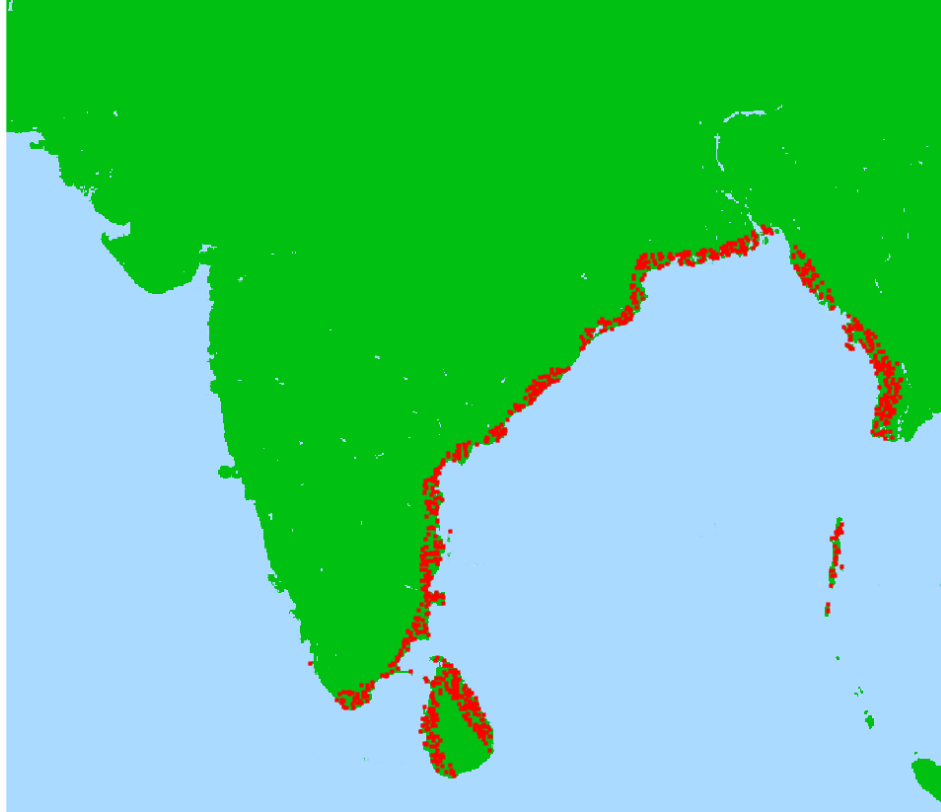


Figure 6: Map with measurement stations.

b

Here, we built a distance matrix from the time-series data. After computing the distances as the 2-norm of the difference between two time-series vectors, we isolated the results obtained for the stations 102, 280, and 10. We obtained that the time-series distance between station 102 and 280 is 3.55, between station 102 and 10 is 4.61, and between station 280 and 10 is 4.84. These distances look fairly

reasonable, since from the plot provided in the assignment text, it can be clearly seen that the station 102 and 280 are closer than 10. Therefore, we these results look consistent with the actual distances.

c

An adjacency matrix using kNN with $k = 3$ in the *or* version was here constructed. From the graph obtained, humans could clearly identify two main clusters on the left-hand side and right-hand side of the graph. In addition to that, from this initial separation, further clusters more weakly separated can be identified on the left.

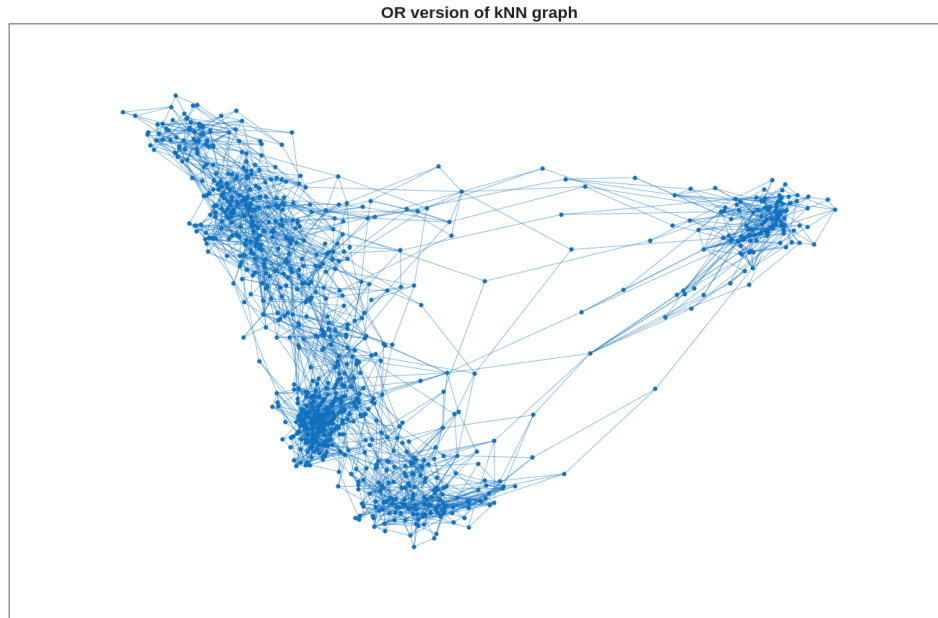
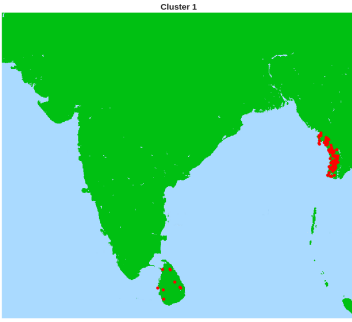


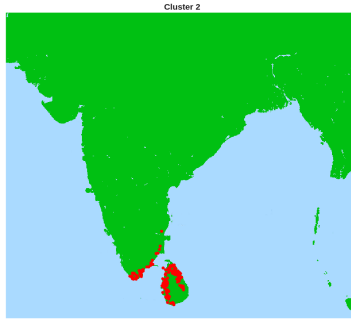
Figure 7: Clusters deriving by kNN graph (OR version).

d

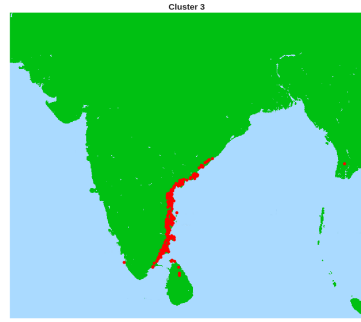
Throughout this task, spectral clustering with seven clusters has been performed based on an unnormalized graph Laplacian. Seven images are displayed below. Each of them show the corresponding cluster on the map.



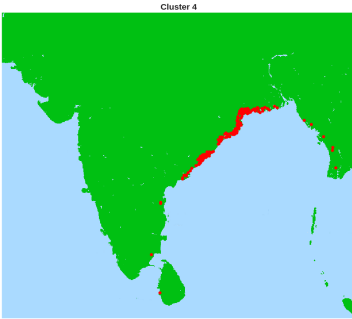
(a) Cluster 1



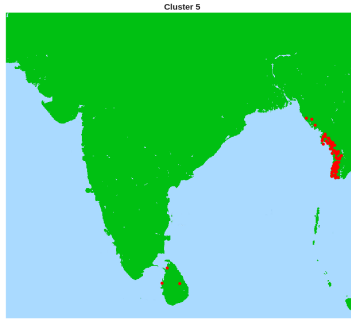
(b) Cluster 2



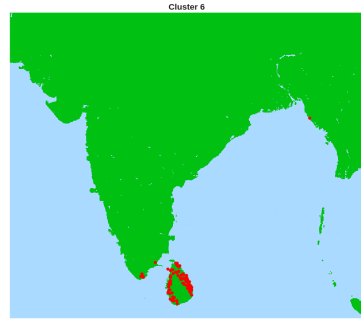
(c) Cluster 3



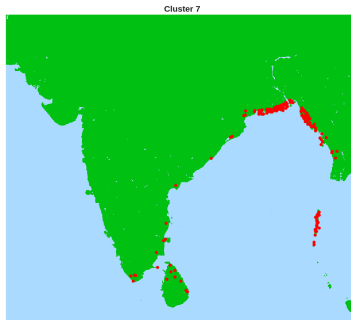
(d) Cluster 4



(e) Cluster 5



(f) Cluster 6



(g) Cluster 7

e

In this task, we present the results that could be obtained by setting $k = 2$ when performing kNN graph. Below, the corresponding graph is presented together with the maps with the seven clusters.

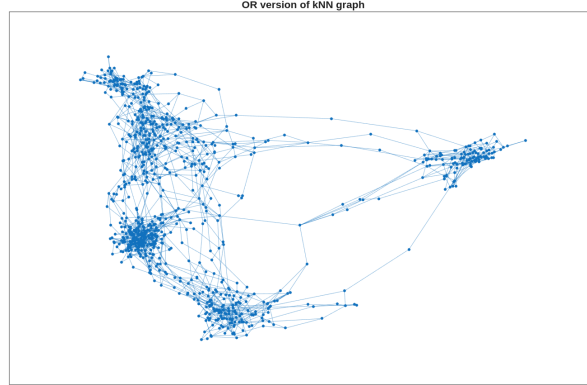
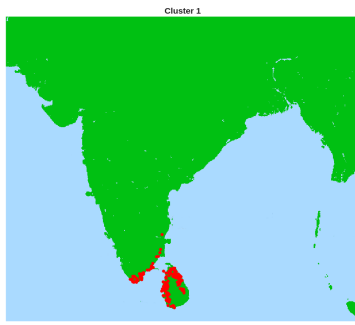
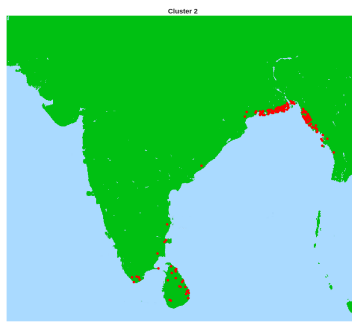


Figure 9: Clusters by kNN graph with $k=2$

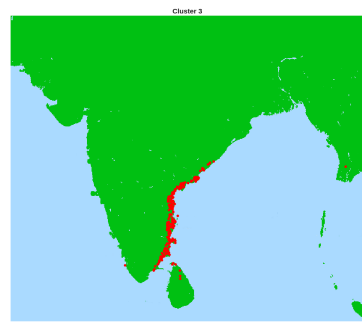
Finally, one can notice that even if the clusters when $k = 2$ are slightly different from the clusters generated when $k = 3$, the general conclusion of the most similar areas for plastic pollution in the map are pretty consistent over the two scenarios. In contrast, what differ the most is the kNN graph, from which humans might be able to clearly identify four highly separated clusters.



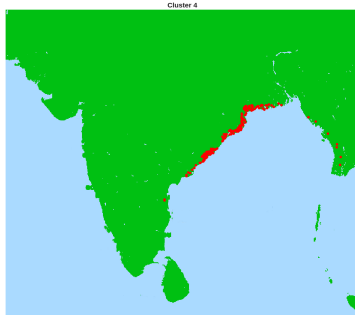
(a) Cluster 1



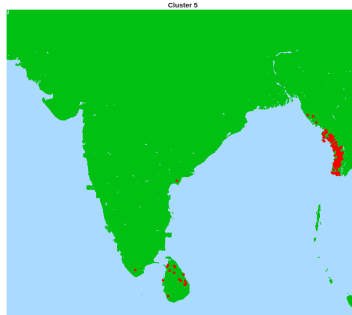
(b) Cluster 2



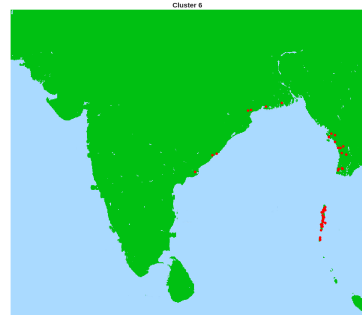
(c) Cluster 3



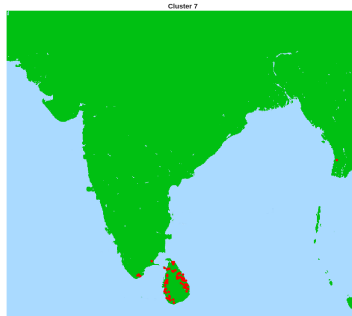
(d) Cluster 4



(e) Cluster 5



(f) Cluster 6



(g) Cluster 7

Exercise 4

a

To identify the two types of items in the zalando dataset, we first plot them randomly for 18 times. The sample we chose are as below:

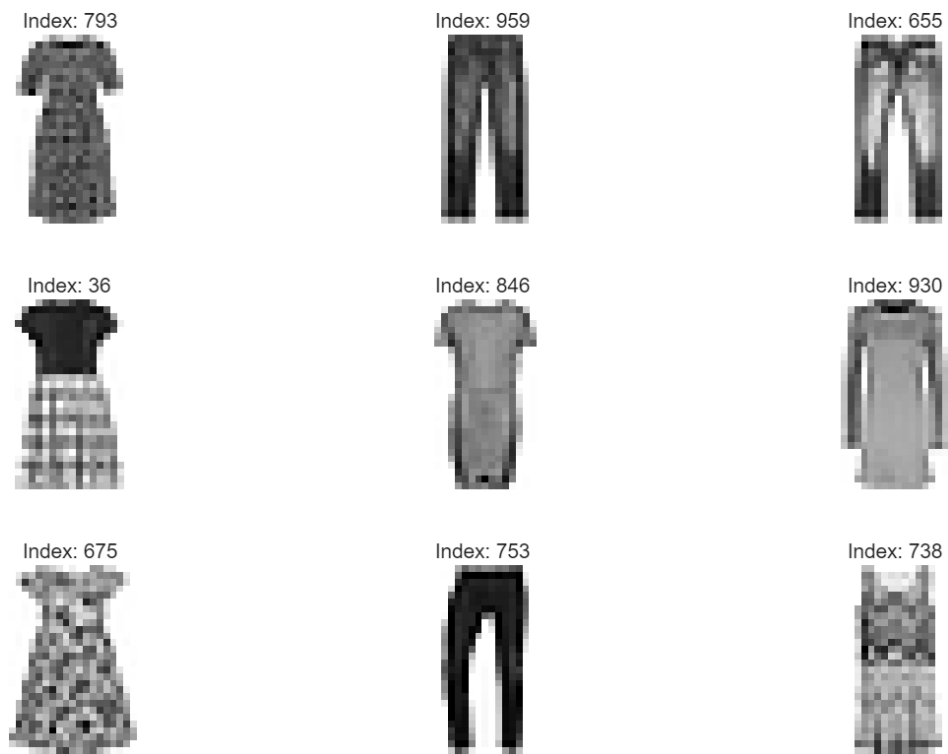


Figure 11: Random pics of items



Figure 12: Random pics of items

According to the information from raw data, there are 9 labels of this dataset and they are We can recognize that the two types are 'Trouser' and 'Dress'.

Label	Description
0	T-shirt/top
1	Trouser
2	Pullover
3	Dress
4	Coat
5	Sandal
6	Shirt
7	Sneaker
8	Bag
9	Ankle boot

b

The four requested figures correspond to the first column image, its closest neighbor, the second column image, and its closest neighbor.

Taking the image most similar to the first column as an example, we examine the distance matrix D constructed from Euclidean distances and locate the smallest nonzero entry d_{j1} in the first column. The corresponding index j identifies the nearest image.

The four resulting images are presented in groups for comparison in Fig 13.

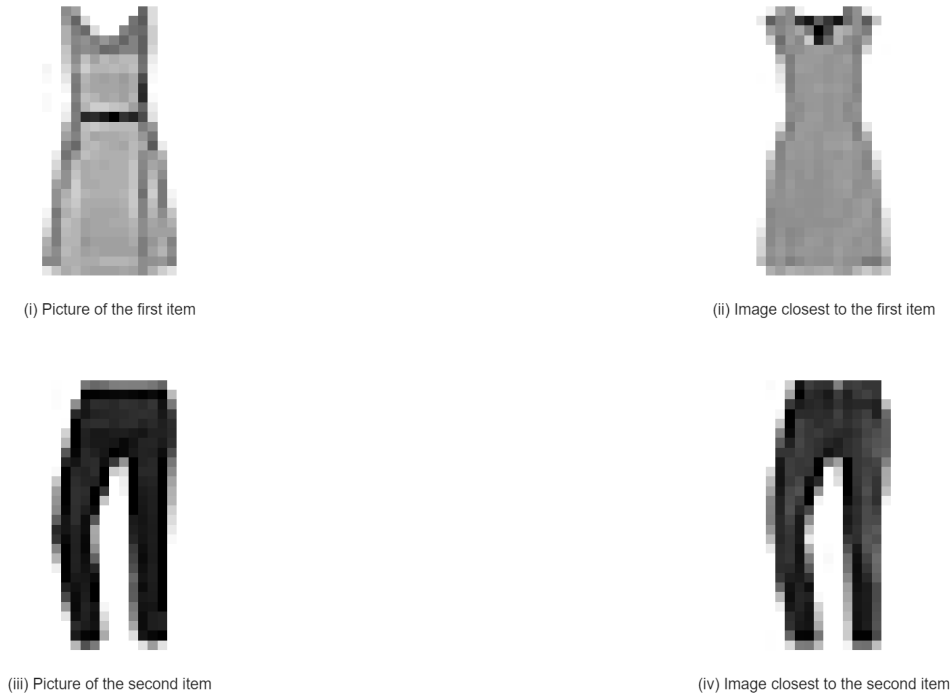


Figure 13: Four images required

c

To implement spectral clustering, we construct the weighted adjacency matrix, compute the degree matrix and obtain the Laplacian matrix. According to the theoretical properties of the Laplacian, the eigenvector corresponding to the second smallest eigenvalue can be used to determine the two clusters via the signs of its entries. Therefore, we first plot the second eigenvector, i.e., the Fiedler vector.

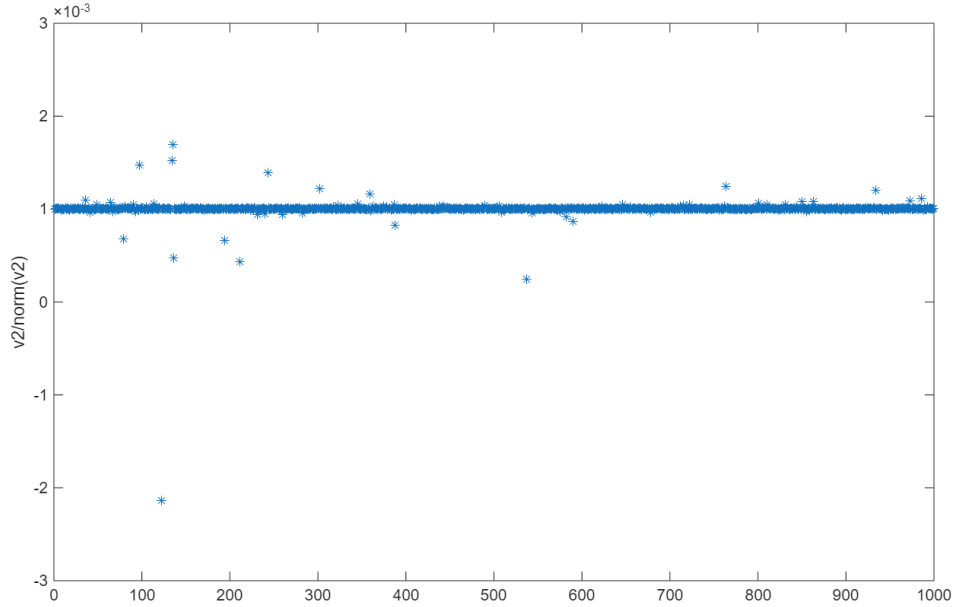


Figure 14: The second eigenvector

As shown in Fig. 14, the majority of the entries of the Fiedler vector are concentrated very closely near zero, with only a small number lying in the intervals $(-0.003, 0)$ and $(0, 0.002)$.

Ideally, a clear separation to two discrete values of Fiedler vector would indicate two well-defined clusters. However, the strong concentration around zero implies weak structural separation in the graph, suggesting that spectral clustering does not work well in our case.

d

Since classification based solely on the sign of the Fiedler vector does not yield a clear separation, and given the prior knowledge that there are 500 images in each class, we instead use the median of the second eigenvector v_2 as the threshold for classification. Specifically, entries smaller than the median are assigned to one cluster, while those larger than the median are assigned to the other, resulting in an index vector idx_2 .

We then compare the predicting label of idx_2 with the ground-truth label vector 'correct' given in the raw dataset to compute the classification accuracy.

By this "sign classification method", we ultimately got 182 items that is wrongly labeled, thus the percentage of correctly classification items is 81.80 %.

e

In this subquestion, we complete and implement the function "hw2_plot_bad_skeleton" to visualize four images that were incorrectly classified by our previous method in the first row. In addition, for each misclassified image, we identify its nearest neighbor by computing their distances and obtain the corresponding index and distance value. These nearest images, together with their distances, are displayed in the second row.

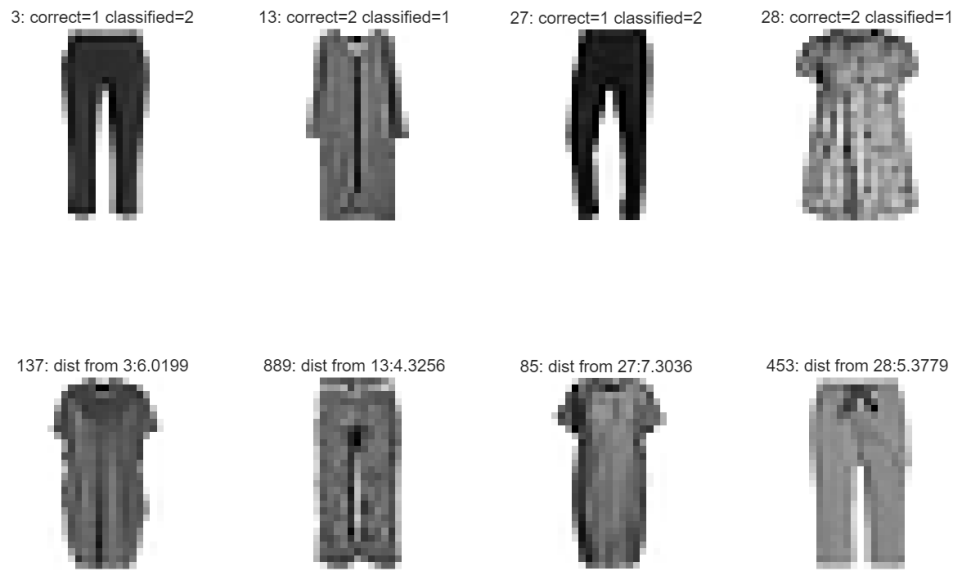


Figure 15: Four incorrectly classified items and their nearest neighbour of different class

The missing part of function "hw2_plot_bad_skeleton" can be complemented as

```

for j=1:plotonly
i=II(j);
% ** Add code here to compute k: the image closest to item(:,i)
% but correct(i) ~= correct(k)
another_ind = find(correct ~= correct(i));
distances_to_others = D(another_ind, i);
[min_dist, local_idx] = min(distances_to_others);
k = another_ind(local_idx);

subplot(2,plotonly,j+4);
zalando_plot(items(:,k));
title([num2str(k), ': dist from ', num2str(i), ': ', num2str(min_dist)]);
end

```

f



Figure 16: zalando_plot(w_weight)

The picture shown in Fig.16 corresponds to the vertical central band of the image. It consists of two grey vertical parts separated by a black region. Since rows 12-16 lie approximately in the middle of the total 28 rows and the "zalando_plot" function applies a transpose A' before displaying the image, these rows turn into middle columns in the visual plot.

By observing images of items and their most similar but in another class items as in Fig.15), we find that the central part often includes important features: for trousers, there are always features like zippers and gap between legs in the vertical middle part; while for dresses, these areas are always filled and contain no blank area. Assigning greater weight to this region emphasizes these distinctive differences, thus increasing the contrast between two classes during the similarity measure process. As a result, this weight strategy helps to improve the effectiveness of spectral clustering, leading to better classification performance.

After applying this weighting strategy, there are only 12 items out of 1000 that are misclassified, leading to the percentage of correctly classification items rises from 81.80 % to 98.80 %.

Answer to the optional question:

We can get better classification by changing x to a bigger number moderately, furthering stress the differences between the middle part and the side part. For example, through testing, when $x=20$, the corresponding accuracy will increase to 99.4 % with only 6 items being misclassified. It's noticeable that not the larger x is, the greater the classification performance be. In addition, multiplying the all-ones weight matrix W_weight by a coefficient smaller than 1 can also improve the classification performance. The underlying idea is similar: while keeping the central weighted region unchanged (x being fixed), reducing the weights of the side areas effectively broadens the difference too, leading to improved clustering results.

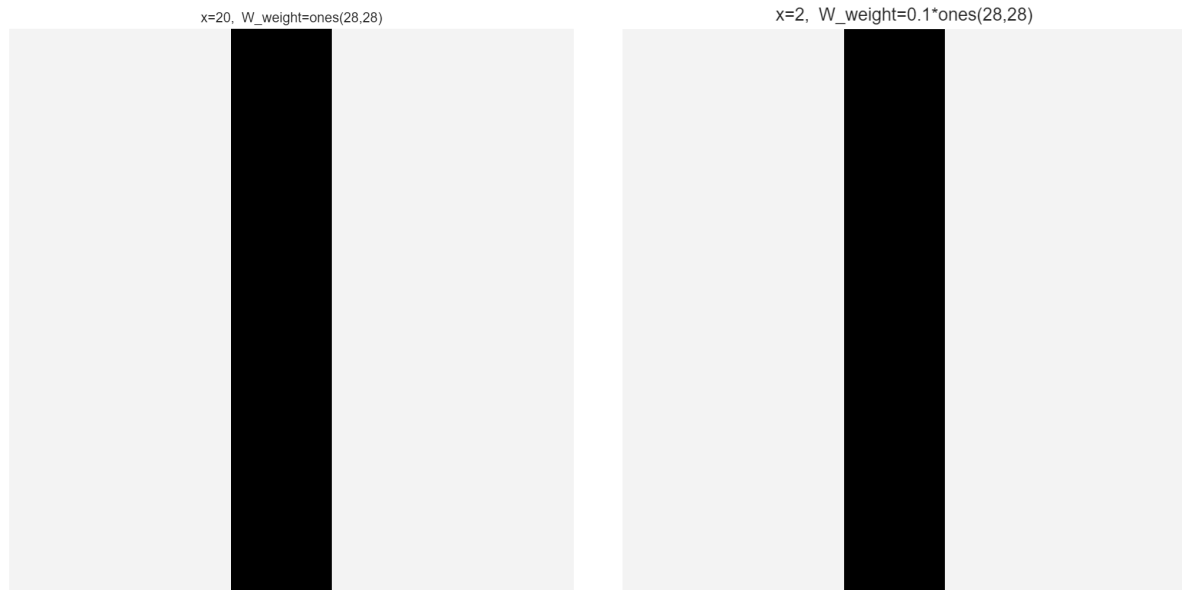


Figure 17: w_weight with different settings

Comparing Fig.16 with Fig.17, it can be clearly observed that, in Figure 17, the contrast between the central and side regions in both subfigures is more distinct compared to Figure 16.

Exercise 5

- **Video quiz 6:** This video is mainly about eigenvalue multiplicity. It first defines the eigenvalue problem, algebraic multiplicity, and geometric multiplicity. Then it gave two lemmas, one is that when matrix A is symmetric, the algebraic multiplicity equals to the geometric multiplicity; Another is that, the linear combination of eigenvectors constitutes another eigenvector. Also, it teaches us to get eigenvalue and eigenvectors in matlab by using command "eig(A)".

- **Video quiz 7a:** This video is about mincut, a method to do clustering. It first defines "cut" with the formula

$$cut(A_1, \dots, A_k) = \frac{1}{2} \sum_{i=1}^k \omega(A_i, \overline{A_i})$$

and gave two examples. Then mincut is defined as

$$\min_{A_1, \dots, A_k} cut(A_1, \dots, A_k).$$

Sometimes, the smallest $cut(A, \overline{A})$ can not give the optimal solution, we may need to modify slightly to find the smallest $Ratiocut(A, \overline{A})$.

- **Video quiz 7b:** This video introduces the Rayleigh-Ritz theorem by first presenting Rayleigh quotient $r(x) = \frac{x^T A x}{x^T x}$. The theorem demonstrates that for a symmetric matrix A , λ_i can be obtained by minimizing $r(x)$, and the bigger i gets, more orthogonal restrictions are required for λ_i , s.t.

$$v_1^T x = 0, v_2^T x = 0, \dots, v_{i-1}^T x = 0$$

. Two examples are then shown to support this theorem.

- **Video quiz 8:** This video focuses on derivatives of eigenvalues and eigenvectors adopting a well-structured approach: it first states the main results, then demonstrates how these results can be applied through examples, and finally provides their detailed derivations. The detailed result is that for a parameter dependent eigenvalue problem $(A(\varepsilon) = A_0 + \varepsilon A_1)$, which satisfies the eigenvalue equation: $A(\varepsilon)x(\varepsilon) = \lambda(\varepsilon)x(\varepsilon)$, we assume that the eigenvalues and eigenvectors admit the following first-order asymptotic expansions:

$$\begin{aligned} x(\varepsilon) &= x_0 + \varepsilon x'(0) + \mathcal{O}(\varepsilon^2), \\ \lambda(\varepsilon) &= \lambda_0 + \varepsilon \lambda'(0) + \mathcal{O}(\varepsilon^2). \end{aligned}$$

Thus, by differentiating at $\varepsilon = 0$, the derivatives satisfy the following linear system:

$$\begin{bmatrix} x'(0) \\ \lambda'(0) \end{bmatrix} = - \begin{bmatrix} A_0 - \lambda_0 I & -x_0 \\ 2x_0^T & 0 \end{bmatrix}^{-1} \begin{bmatrix} A_1 x_0 \\ 0 \end{bmatrix},$$

where $x_0 = x(0)$ and $\lambda_0 = \lambda(0)$.

- **Video quiz 9:** This video presents the proof for Reighleig-Ritz theorem. The theorem states that the k -th eigen-value (λ_k) corresponds to the minimum of the Reighleig-Ritz coefficient, i.e. $\lambda_k = \min_{x \in D_k(A)} r_A(x)$, where $D_k(A)$ is the set of orthogonal vectors s.t. $D_k(A) = \{x \in \mathbb{R}^n : \|x\| = 1 \wedge v_1^T x = \dots = v_n^T x = 0\}$. The proof is developed in two parts a and b. On the one hand part a is used to show that we can express the theorem with diagonal matrices without loss of generality obtaining that $\min_{x \in D_k(A)} r_A(x) = \min_{z \in D_k(\Lambda)} r_\Lambda(z)$, and the Reighleigh-Ritz coefficient of a diagonal matrix expressible as $r_\Lambda(z) = z_1^T \lambda_1 + \dots + z_n^T \lambda_n$. On the other hand, part b makes use of specific structures and characteristics of the theorem to show that both $\min_{z \in D_k(\Lambda)} r_\Lambda(z) \leq \lambda_k$ and $\min_{z \in D_k(\Lambda)} r_\Lambda(z) \geq \lambda_k$ are true.